Sungwoo Kim

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Software Engineer (System Security)

Interests: Operating Systems Security, Embedded Systems, Device Firmware, Kernel Drivers, Vulnerability Research

Conducted Linux kernel security research, using Syzkaller and KCOV to discover previously unknown kernel vulnerabilities. Investigated memory corruption bugs, including use-after-free and out-of-bounds access, and evaluated kernel hardening mechanisms. Developed fuzzing techniques to improve vulnerability discovery in operating system components.

Education

Ph.D. in Computer Science, Purdue University **2021 Aug. – May 2027 (Expected)**
Systems security
B.S. in Engineering, Kwangwoon University **Mar. 2014 – Feb. 2020**
Electronics and communications engineering

Publications

4. **SoK: Kicking CAN Down the Road Systematizing CAN Security Knowledge**; Khaled Serag, Zhaozhou Tang, Sungwoo Kim, Vireshwar Kumar, Dave (Jing) Tian, Saman Zonouz, Raheem Beyah, Berkay Celik, and Dongyan Xu; *34th ISOC Network and Distributed System Security Symposium (NDSS'27)*; *acceptance rate: TBD*

3. **FuzzBT: Holistic-State-Guided Fuzzing for Bluetooth Host Stack in Kernels**; Sungwoo Kim, Hui Peng, Imtiaz Karim, Ruoyu Wu, Jianliang Wu, Elisa Bertino, Mathias Payer, and Dave (Jing) Tian; *USENIX WOOT Conference on Offensive Technologies (WOOT '26)*; *acceptance rate: 33%*

2. **Fuzz The Power: Dual-role State Guided Black-box Fuzzing for USB Power Delivery**; Kyungtae Kim, Sungwoo Kim, Kevin Buttler, Antonio Bianchi, Rick Kennell, Dave (Jing) Tian; *32nd USENIX Security Symposium (Security'23), Anaheim, CA, August 2023 (USENIX '23)*; *Acceptance Rate: 29.2%*

1. **ShadowAuth: Backward-Compatible Automatic CAN Authentication for Legacy ECUs**; Sungwoo Kim, Gisu Yeo, Taegyu Kim, Junghwan "John" Rhee, Yuseok Jeon, Antonio Bianchi, Dongyan Xu, and Dave (Jing) Tian; *Proceedings of the 2022 ACM Asia Conference on Computer and Communications Security (ASIACCS '22)*; *acceptance rate: 18.4%*

Vulnerability Findings

Linux Kernel

- Bluetooth: 9 CVEs [1]
- NVMe: 3 CVEs [2]
- CXL: 1 CVE [3]

Apple XNU kernel

- 5 Bugs. Writeup: https://sung-woo.kim/b/syz-xnu/bugs.html

Embedded system

- Drone (PX4) : 1 CVE [4]

Misc.

- Reviewing service: TDSC'26 and SmartSP'26
- Architecture Evaluation Committee (AEC): USENIX'24 and '26.
- Built and maintain a search engine, cspapers.org (200+ GitHub stars, 120 DAU)

[1] CVE-2024-50255, CVE-2024-38620, CVE-2024-36968, CVE-2024-36013, CVE-2024-36012, CVE-2024-36011, CVE-2023-40283, CVE-2023-28866, CVE-2022-45934

[2] CVE-2026-23244, CVE-2026-43448, CVE-2026-43449

[3] CVE-2026-74272

[4] CVE-2021-34125